

Falah Sheikh

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Education

- Expected 2028 **University of Calgary**, M.Sc. in Computer Science.
Advisor: Fateme Rajabiyazdi
- 2022 – 2026 **University of Calgary**, B.Sc. in Computer Science.

Research Experience

- Aug 2026 – present **HealthVisFutures Lab, Interactions Lab**
Graduate Research Assistant, University of Calgary
Advisor: Fateme Rajabiyazdi
- May 2024 – May 2026 **Data and Network Sciences & Applications Lab**
Undergraduate Research Assistant, University of Calgary
Advisors: Jon Rokne, Reda Alhajj
- *Prompt structure and LLM hallucination* (Sep 2025 – May 2026). Measured hallucination across four commercial language models and six prompt categories on 7,150 scored responses, separating prompt structure from model-level susceptibility (up to 78.00% on sentence-completion prompts).
 - *Explainable Alzheimer's detection from structural MRI* (Jan – Sep 2025). Benchmarked lightweight models (MobileNetV2, EfficientNetV2B0) to classify coronal MRI slices from 102 ADNI subjects as cognitively normal, early MCI, or late MCI; EfficientNetV2B0 reached 88.0% mean accuracy in 5-fold cross-validation, and Grad-CAM++ maps highlighted regions commonly associated with early-stage degradation.
 - *Lunar age estimation from images* (May 2024 – Jan 2025). Trained CNNs to predict the day of the synodic month (30 classes) from a single lunar image (best accuracy 82.74%).

Publications

Journal Articles

- F. Sheikh, A. Al Marouf, J. Rokne, R. Alhajj. Lightweight deep learning models with explainable AI for early Alzheimer's detection from standard MRI scans. *Diagnostics* 15(21), 2709 (2025). doi:[10.3390/diagnostics15212709](https://doi.org/10.3390/diagnostics15212709)
- Z. Nissar, F. Sheikh, A. Al Marouf, J. Rokne, R. Alhajj. Automated lunar age detection through determining the day of the synodic month using convolutional neural networks. *Scientific Reports* 15, 35973 (2025). doi:[10.1038/s41598-025-18165-w](https://doi.org/10.1038/s41598-025-18165-w)

Conference Proceedings

- F. Sheikh, A. Al Marouf, R. Alhajj, J. Rokne. NeuroXAI: Lightweight explainable early Alzheimer's detection for resource-constrained clinical settings. *Medical Imaging 2026: Imaging Informatics*, Proc. SPIE 13930, 139301R, Vancouver, Canada (2026). doi:[10.1117/12.3088200](https://doi.org/10.1117/12.3088200)

Awards & Honors

- Nov 2025 Outstanding research presentation, Community Mental Health and Well-Being Strategy Award (CA\$1,000).
- May 2025 Alberta Innovates Summer Research Studentship (CA\$7,500).

Teaching

- Fall 2026 Teaching Assistant, CPSC 217: Introduction to Computer Science for Multidisciplinary Studies I, *University of Calgary*